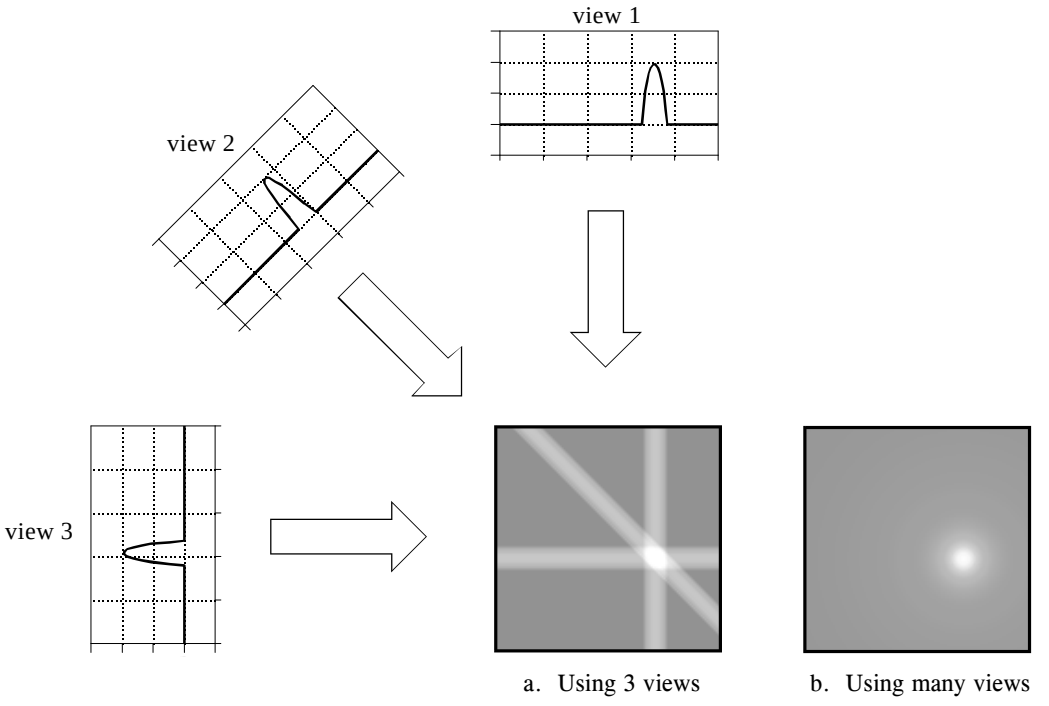

FIGURE 25-16
Backprojection. Backprojection reconstructs an image by taking each view and *smearing* it along the path it was originally acquired. The resulting image is a blurry version of the correct image.

technique, called **backprojection** or **simple backprojection.** Figure 25-16 shows that simple backprojection is a common sense approach, but very unsophisticated. An individual sample is backprojected by setting all the image pixels along the ray pointing to the sample to the same value. In less technical terms, a backprojection is formed by *smearing* each view back through the image in the direction it was originally acquired. The final backprojected image is then taken as the sum of all the backprojected views.

While backprojection is conceptually simple, it does not correctly solve the problem. As shown in (b), a backprojected image is very *blurry*. A single point in the *true* image is reconstructed as a circular region that decreases in intensity away from the center. In more formal terms, the point spread function of backprojection is circularly symmetric, and decreases as the reciprocal of its radius.

Filtered backprojection is a technique to correct the blurring encountered in simple backprojection. As illustrated in Fig. 25-17, each view is *filtered* before the backprojection to counteract the blurring PSF. That is, each of the one-dimensional views is convolved with a one-dimensional filter kernel to create a set of *filtered views*. These filtered views are then backprojected to provide the reconstructed image, a close approximation to the "correct" image. In fact, the image produced by filtered backprojection is *identical*